LookInsight — Insider Conviction Signals

A Methodology and Performance Brief

Snapshot date	April 19, 2026
Cohort size	141 mature strong_buy signals
Universe	US-listed equities via SEC EDGAR Form 4
Delivery scope	Institutional — quantitative hedge fund research desks
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Table of Contents

1. Executive Summary . . . 3

2. Thesis . . . 3

3. The Funnel . . . 4

4. Methodology . . . 5

- 4.1 Why P transactions only . . . 5
- 4.2 Why 2+ distinct insiders . . . 5
- 4.3 Why $100K+ cluster value . . . 6
- 4.4 Why midcap $300M – $5B . . . 6
- 4.5 Why earnings proximity ≤ 60 days . . . 6
- 4.6 Why filing-date returns (not transaction-date returns) . . . 7
- 4.7 What we tested and rejected . . . 7

5. Performance . . . 7

- 5.1 Headline metrics . . . 8
- 5.2 Return distribution . . . 8
- 5.3 Signal-level alpha . . . 9
- 5.4 Breakdown by number of insiders . . . 9
- 5.5 High-conviction subset (signal_level) . . . 10
- 5.6 Hostile activist overlap . . . 10
- 5.7 Signal-level performance log . . . 10

6. Academic Foundation . . . 17

7. Appendix . . . 18

- File index . . . 18
- Glossary . . . 18

1. Executive Summary

LookInsight produces a filtered alternative-data feed of high-conviction insider-buying signals. This brief documents the methodology that compresses ~67,000 raw SEC Form 4 transactions down to 141 mature, high-quality signal clusters — and presents the performance evidence on that cohort.

Headline results (cohort of 141 mature strong_buy signals):

Metric	Value	95% Confidence Interval	p-value
Hit rate (90-day return > 0)	67.4%	[59.3%, 74.6%] (Wilson)	<0.001 vs 50% null
Mean alpha vs SPY (90d)	+9.0 pp	[+3.3, +14.6] pp (t-dist)	0.0022
Mean 90-day return	+14.2%	—	—
Mean 90-day SPY return	+5.3%	—	—
Positive-alpha signals	86 / 141	—	—

The thesis in one sentence: raw SEC Form 4 filings are ~99% noise, and the filter is the product. LookInsight's methodology funnel compresses ~67,000 insider transactions into 141 high-conviction clusters, and that compression is where the measured alpha lives.

What this brief is not. This is a signal-quality brief, not a portfolio backtest. Strategy statistics — Sharpe ratio, drawdown, turnover, capacity, hold-period sensitivity — depend on the construction framework a desk chooses and are deliberately left to the buyer. LookInsight delivers the signal; the desk benchmarks the strategy built on it.

Section 3 walks through the funnel. Section 4 explains each filter rule. Section 5 presents the performance evidence and breakdowns. Section 6 sets out the academic foundation. Section 7 indexes the supporting files and glossary.

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2. Thesis

Insider behavior — specifically, open-market purchases by officers and directors — is a well-documented alpha source. Three decades of peer-reviewed research (see Section 6) establish the signal's foundation. The open question for a hedge fund research desk is not *whether* insider purchases predict returns, but *how to extract them at production scale* without being buried by the noise of RSU vesting, 10b5-1 execution, option mechanics, and structured placements.

LookInsight's position. Insider behavior alone — pure signal, no cross-filing overlays, no alternative-data fusion, no macro factors — produces statistically significant alpha over SPY when the right filters are applied to SEC Form 4 at scale. This brief documents the filters, the performance, and the audit path an analyst can follow to verify every row.

What this saves a research desk. Months of SEC filing parsing, transaction-type classification, LLM-based disambiguation of genuine open-market purchases from placements, historical market cap estimation, earnings-proximity gating, maturity accounting, and statistical validation — compressed into a 33-column data appendix consumable directly in research workflows.

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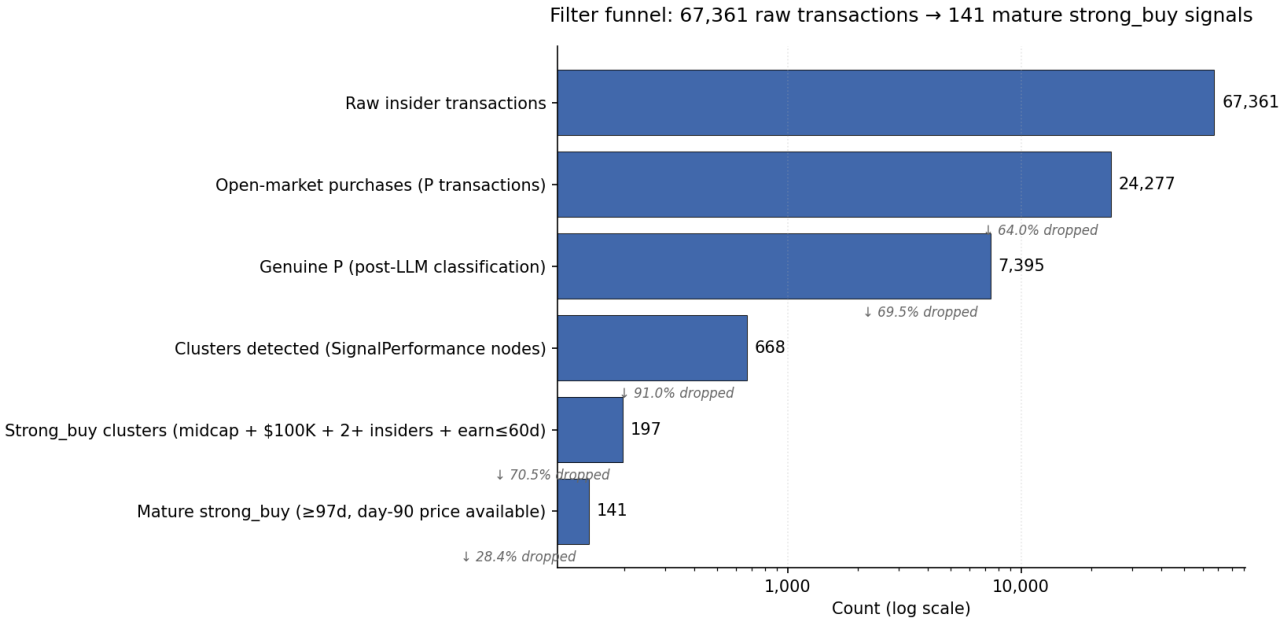
3. The Funnel

Insider signals are noise by default. The product is the filter.

Across 22 months of SEC Form 4 ingestion, LookInsight's pipeline compresses **67,361 raw transactions** into **141 mature strong_buy signals**.



Stage-by-stage:

1. **Raw insider transactions (67,361).** All Form 4 line items ingested: purchases, sells, option exercises, gifts, derivative events, everything.
2. **Open-market purchases (24,277).** Filter to `transaction_code = 'P'`. Drops option exercises (M, C, W), sells (S, F, G, D), gifts, and derivative events. **64% drop.**
3. **Genuine P transactions (7,395).** An LLM-based classifier (Claude Haiku 4.5 plus a 21-rule prefilter) removes structured placements, RSU vestings reported as P, director option grants, employee stock-purchase programs, and 5+ buyers at identical price (typical IPO allocations and private placements). **70% drop.**

4. **Clusters detected (668).** Trades grouped into rolling-window clusters by issuer, deduplicated across multi-entity filings, and aggregated by distinct insider name. This is the full SignalPerformance node space — all directions, all conviction tiers, all maturity states. **91% drop** — the single largest compression step.
5. **Strong_buy clusters (197).** Apply cohort constraints: direction = buy, ≥2 distinct insiders, ≥$100K aggregate cluster value, historical market cap in the $300M–$5B midcap band, earnings proximity ≤60 days. **71% drop.**
6. **Mature strong_buy (141).** Restrict to signals ≥97 days old with an observed day-90 closing price. **28% drop** — remaining signals are recent filings still within the maturity window.

Compression ratio: roughly 477 raw transactions yield 1 mature strong_buy signal. The filter discards **99.8%** of raw input.

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4. Methodology

Each filter in the funnel exists for a specific, evidence-backed reason. This section walks through them.

4.1 Why P transactions only

Of the thirteen SEC transaction codes, only `P` (open-market purchase) is retained. Sells (`S`), forced sells (`F`), option exercises (`M`, `C`, `W`), and gifts (`G`, `D`) are excluded because they carry substantial non-conviction motivation: tax withholding, diversification, scheduled 10b5-1 execution, or derivative mechanics unrelated to a directional view. Buying conviction is cleaner.

This asymmetry is the central empirical finding of **Jeng, Metrick & Zeckhauser (2003, *Review of Financial Studies*)**: insider purchases produce approximately 11% in abnormal annual returns, while insider sales show no comparable predictive power. **Cohen, Malloy & Pomorski (2012, *Journal of Finance*, "Decoding Inside Information")** further isolate *opportunistic* open-market purchases — trades that do not follow a regular annual pattern — as the high-signal subset, reinforcing the filter to genuine P and the engineering work of stripping out scheduled / routine purchases. Internal performance testing confirmed that aggregating across all bullish transaction codes diluted hit rate without meaningfully improving breadth.

4.2 Why 2+ distinct insiders

Single-insider buying is a high false-positive regime. A lone director putting $100K on the tape may be signaling, hedging a personal position, or complying with an implicit officer share-ownership guideline. Two or more distinct insiders buying within a rolling window demonstrate shared conviction — the probability that two officers independently reach the same bullish view without real information is low. This is the anchor of the cluster construct.

Classical insider-alpha studies are single-insider by default (**Lakonishok & Lee 2001**; **Jeng, Metrick & Zeckhauser 2003**), which understates the signal strength available from clustered trading. The multi-insider extension treats distinct-insider consensus as the dispositive filter — an engineering refinement on top of the academic baseline, consistent with the broader behavioral-consensus literature that correlated trades carry more informational content than solitary ones.

4.3 Why $100K+ cluster value

Below this threshold, cluster dollars are dominated by token show-of-faith trades rather than meaningful capital commitment. The threshold selects for insiders putting enough of their own capital on the line that the trade carries economic weight.

Seyhun (1986, *Journal of Financial Economics*, "Insiders' profits, costs of trading, and market efficiency") documented that larger-dollar insider trades carry disproportionately greater informational content, establishing the academic basis for applying a cluster-value floor. The $100K threshold is calibrated to midcap share prices and typical insider position sizes — low enough to retain breadth, high enough to discard token trades.

4.4 Why midcap $300M – $5B

The size effect in insider-purchase alpha is well-documented. **Lakonishok & Lee (2001, *Review of Financial Studies*)** found abnormal returns following insider purchases to be substantially stronger in smaller firms, and **Jeng, Metrick & Zeckhauser (2003)** replicated this size tilt in their return estimation. The academic baseline supports a small-to-mid-cap focus.

Our implementation sharpens the band empirically: the $5B–$10B bucket produced a hit rate of 38.1% versus 67.4% for issuers under $5B ($p = 0.018$ — confidence intervals do not overlap). The upper cap was therefore tightened from $10B to $5B. The $300M floor excludes microcaps where single-trade price manipulation risk is higher and liquidity is thinner for systematic execution.

A note on in-sample calibration. The $5B cap is the one filter parameter empirically tuned on the current data window; every other filter — P-only, 2+ distinct insiders, $100K cluster-value floor, ≤60-day earnings proximity, filing-date return anchor — is pre-specified from the cited literature rather than selected in-sample. A desk evaluating the forward profile of this cohort should expect hit rate to settle in the low-60s rather than the in-sample 67.4%, consistent with a modest shrinkage on the one parameter we fit. Even under that shrinkage, the alpha case remains literature-backed and statistically significant.

4.5 Why earnings proximity ≤ 60 days

Clusters formed within 60 days of the issuer's next earnings release produce significantly better results ($p = 0.003$ on this single rule alone). This is an information-asymmetry window: insiders likely have forward visibility on revenue, margin, or guidance that the market has not priced yet.

Ke, Huddart & Petroni (2003, *Journal of Accounting and Economics*, "What insiders know about future earnings and how they use it: Evidence from insider trades") showed that insider trades cluster in advance of material earnings outcomes, and that the predictive content is concentrated in the months leading up to announcement — the 60-day window is designed to capture this channel. Stacking multiple weak rules on top of this one cost signals for less than one percentage point of hit-rate improvement during empirical testing, so the methodology retains a single, evidence-based temporal rule.

4.6 Why filing-date returns (not transaction-date returns)

Returns are anchored to the **filing date**, not the transaction date. The 1–3 day gap between trading and SEC filing is invisible to an external reader — measuring returns from the transaction date embeds a look-ahead bias. Filing date is the earliest moment a third party could observe and act on the signal, and is therefore the only defensible anchor for alpha claims.

The filing-date anchor is the methodological convention in the return-estimation literature (**Lakonishok & Lee 2001**; subsequent work including **Cohen, Malloy & Pomorski 2012** frame their analysis around observable filing events rather than unobserved transaction dates). Using transaction date inflates the measured return by the size of the market's move during the reporting gap — a one-to-three-day phantom alpha that does not exist for a trader acting on the public filing.

4.7 What we tested and rejected

The following hypotheses were evaluated during methodology development and not promoted into the production filter set:

- **Single-insider dominance as a primary filter** (>70% of cluster value attributable to one insider). Directional evidence only; did not reach statistical significance on a representative sample.
- **Conviction tiers as hard exclusions.** The separation between tiers was directional ($p = 0.11$) but not robust enough to justify dropping signals. Retained as metadata.
- **Hostile activist exclusion as a hard filter.** 88% of a small losing subset carried hostile-activist overlap, but absolute sample size was insufficient to justify a binary filter. Retained as an informational flag (see Section 5.6).
- **Alternative alpha sources** (legislative trading feeds, corporate-action metadata combinations, network-topology features over insider relationships). Each either underperformed insider clusters on measured predictive value or carried reporting-lag constraints incompatible with systematic trading.

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5. Performance

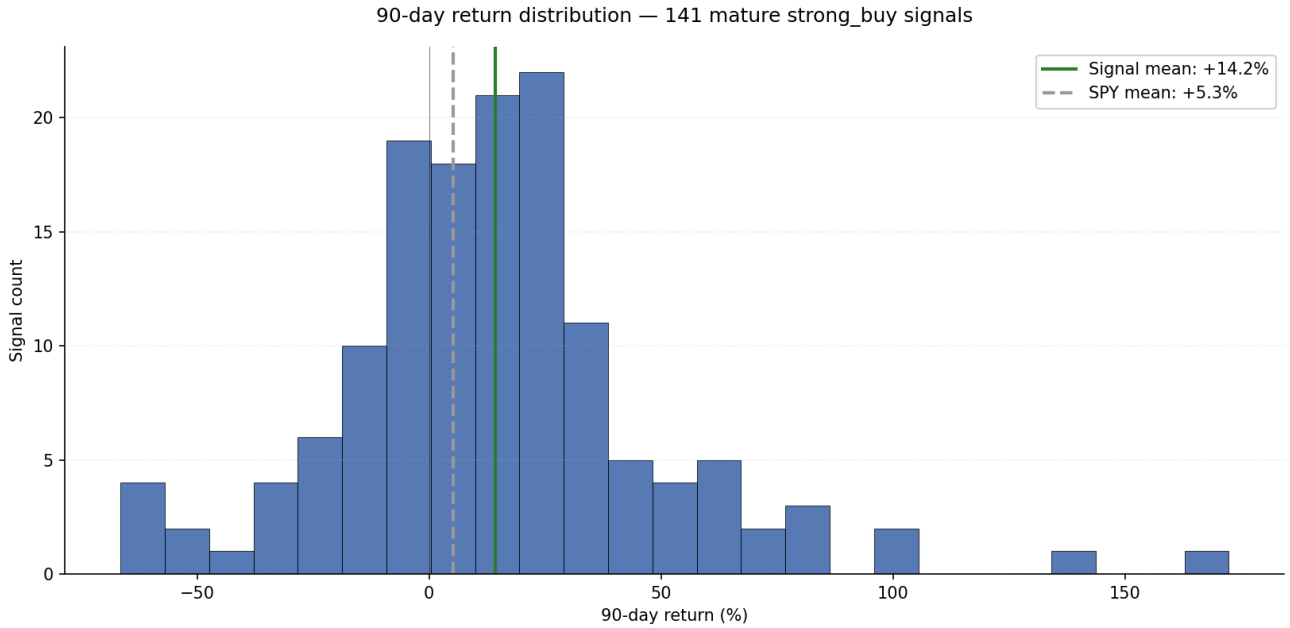
All results are for the 141-signal mature strong_buy cohort. Returns are computed from the filing date, 90 calendar days forward.

5.1 Headline metrics

Metric	Value	95% CI	p-value
Hit rate	67.4%	[59.3%, 74.6%]	<0.001
Mean alpha vs SPY	+9.0 pp	[+3.3, +14.6]	0.0022
Mean 90-day return	+14.2%	—	—
Median 90-day return	+11.4%	—	—
Mean SPY 90-day return	+5.3%	—	—
Positive alpha	86 / 141	—	—
Positive raw return	95 / 141	—	—

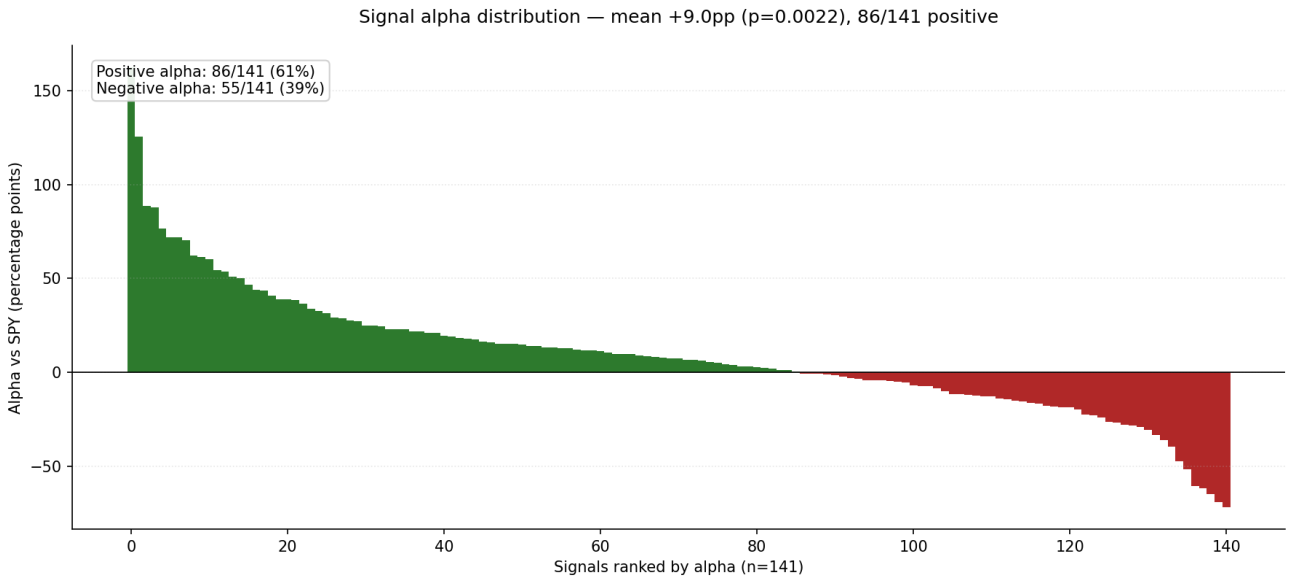
Hit-rate CI is computed via the Wilson score interval; alpha CI via the t-distribution on the sample mean. p-values are two-sided (exact binomial test against 0.5 for hit rate, one-sample t-test against 0 for alpha).

5.2 Return distribution



The distribution has a long right tail — a handful of signals deliver returns above +50%. The signal-mean line sits approximately 9 percentage points above the SPY-mean line for the same 90-day windows. The mass on the positive side of zero reflects the 67% hit rate.

5.3 Signal-level alpha



Ranking each signal by its 90-day alpha makes the distribution explicit: 86 of 141 signals (61%) produce positive alpha vs SPY, with the largest single signal above +140 percentage points and the largest negative outlier near −70. The cohort's mean alpha of +9.0 pp is not driven by a single extreme tail — the bulk of the positive-alpha mass sits between +5 and +40 pp. Note that hit rate (95/141 = 67%, based on raw return > 0) and positive-alpha count (86/141 = 61%) differ because SPY itself advanced during several signal windows; signals that tracked flat would have positive raw return but negative alpha.

5.4 Breakdown by number of insiders

Insiders in cluster	n	Hit rate	Mean alpha (pp)
2	78	65.4%	+8.2
3	26	73.1%	+9.4
4	21	76.2%	+8.3
5+	16	56.3%	+12.6

Two observations. First, hit rates in the 2-, 3-, and 4-insider buckets are all above 65%, which suggests the core cluster hypothesis is stable across cluster sizes rather than being driven by a particular size regime. Second, the 5+ bucket has a lower hit rate but higher mean alpha — the few wins in large clusters are disproportionately large.

The 16 signals in the 5+ bucket span 12 distinct SIC industry codes (biotech, pharma, paper, chemicals, insurance, real estate, retail, mining, services) and carry no hostile-activist overlap, so the lower hit rate is not concentration in a single failing theme. The higher mean alpha is carried by several outsized winners

with individual alpha exceeding +70 pp. We expose the underlying cluster size in the `num_insiders` column; analysts optimizing for hit rate may choose to cap at 4 insiders, while those willing to trade hit rate for alpha magnitude can keep the full cohort.

5.5 High-conviction subset (`signal_level`)

The `signal_level` column in the data appendix splits the cohort into two conviction bands:

- **`high`** — clusters with 3+ distinct insiders, or with 2+ insiders carrying an officer-level majority and cluster value ≥ $200K. **77 signals.**
- **`medium`** — the remaining strong_buy signals (typically 2-insider clusters that do not meet the officer-majority threshold). **64 signals.**

Signal level	n	Hit rate	Mean 90d return	Mean alpha (pp)
high	77	68.8%	+15.1%	+9.8
medium	64	65.6%	+13.2%	+8.0

Both bands are alpha-positive. The `high` band is stronger across all three metrics. Desks building more concentrated strategies may filter to `signal_level = 'high'`; desks optimizing for breadth can use the full cohort. The column is exposed directly in the data appendix so either choice can be made independently.

5.6 Hostile activist overlap

The cohort includes 3 signals (2.1%) where the issuer had a contemporaneous Schedule 13D filing carrying hostile-activist language.

Hostile flag	n	Hit rate	Mean alpha (pp)
False	138	68.1%	+9.2
True	3	33.3%	−2.0

The hostile subset underperforms substantially (1 winner out of 3), directionally consistent with the earlier qualitative finding that hostile-activist overlap is a negative predictor. `hostile_flag` is exported as an informational column — it is optionality the analyst controls, not a claim from LookInsight.

5.7 Signal-level performance log

Signals are grouped below by filing month, most recent first. Each month header shows the aggregate count, hit rate, mean 90-day return, and mean alpha for that month; within the month, rows are individual signals sorted by filing date descending. `Conv` = `signal_level` (HIGH / MEDIUM); `#I` = distinct insiders in the cluster; † marks signals with a contemporaneous hostile Schedule 13D filing.

December 2025

n = 18 · hit rate 72% · mean return +11.9% · mean alpha +12.7 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-12-22	VSTS	HIGH	4	$7.14	$7.72	+8.1%	+12.2
2025-12-18	ANDG	HIGH	5	$25.40	$28.27	+11.3%	+13.2
2025-12-18	SEI	MEDIUM	2	$42.96	$68.56	+59.6%	+61.5
2025-12-16	SMRT	MEDIUM	2	$2.05	$1.72	-16.1%	-14.9
2025-12-16	RLMD	MEDIUM	2	$4.50	$6.18	+37.3%	+38.5
2025-12-16	MLP	MEDIUM	2	$16.81	$16.02	-4.7%	-3.5
2025-12-12	AARD	MEDIUM	2	$14.43	$5.26	-63.5%	-61.5
2025-12-12	CE	HIGH	2	$43.48	$59.60	+37.1%	+39.1
2025-12-09	MNR	MEDIUM	2	$11.74	$13.26	+12.9%	+13.3
2025-12-09	ARX	HIGH	6	$15.37	$10.94	-28.8%	-28.4
2025-12-09	MRVI	MEDIUM	2	$3.69	$3.63	-1.6%	-1.2
2025-12-08	AEBI	HIGH	3	$12.36	$13.50	+9.2%	+9.7
2025-12-08	XRN	HIGH	2	$31.91	$35.52	+11.3%	+11.8
2025-12-04	MSBI	HIGH	4	$19.73	$22.62	+14.7%	+14.2
2025-12-03	ANNX	MEDIUM	2	$4.28	$5.43	+26.9%	+27.1
2025-12-02	ENR	MEDIUM	2	$16.80	$20.50	+22.0%	+21.0
2025-12-01	HLF	MEDIUM	2	$12.63	$19.24	+52.3%	+51.1
2025-12-01	ADTN	HIGH	3	$7.91	$9.96	+25.9%	+24.7

November 2025

n = 17 · hit rate 59% · mean return +1.2% · mean alpha -1.3 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-11-28	RPD	HIGH	4	$15.68	$6.39	-59.2%	-60.4
2025-11-26	CRSR	MEDIUM	2	$6.14	$5.45	-11.2%	-12.7
2025-11-25	EVLV	HIGH	4	$6.20	$4.90	-21.0%	-22.4
2025-11-24	BBWI	HIGH	5	$15.43	$22.52	+46.0%	+43.6
2025-11-24	JACK†	MEDIUM	2	$17.36	$17.07	-1.7%	-4.0
2025-11-24	RXO	HIGH	3	$12.16	$14.66	+20.6%	+18.2
2025-11-21	GLRE	HIGH	3	$13.00	$13.97	+7.5%	+3.3
2025-11-20	CNS	MEDIUM	2	$58.86	$65.38	+11.1%	+5.6
2025-11-14	FRPH	HIGH	3	$24.55	$23.69	-3.5%	-5.2
2025-11-13	ITGR	HIGH	4	$68.05	$87.18	+28.1%	+24.8
2025-11-13	CLVT	MEDIUM	2	$3.50	$1.81	-48.3%	-51.6

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-11-12	FRSH	HIGH	2	$11.64	$8.73	-25.0%	-26.6
2025-11-10	WWW	HIGH	3	$16.15	$18.09	+12.0%	+9.9
2025-11-10	UTZ	HIGH	4	$9.85	$10.86	+10.2%	+8.1
2025-11-06	BBT	HIGH	4	$25.63	$30.20	+17.8%	+15.2
2025-11-04	OBK	HIGH	6	$34.30	$44.03	+28.4%	+25.1
2025-11-03	IRDM	MEDIUM	2	$18.04	$19.65	+8.9%	+6.8

October 2025

n = 2 · hit rate 0% · mean return -10.4% · mean alpha -11.9 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-10-28	ASIC	MEDIUM	2	$18.96	$17.78	-6.2%	-7.3
2025-10-22	ANGO	HIGH	3	$12.14	$10.36	-14.7%	-16.4

September 2025

n = 6 · hit rate 83% · mean return +14.4% · mean alpha +10.1 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-09-30	PGEN	HIGH	4	$3.29	$4.45	+35.3%	+31.7
2025-09-16	ZVRA	HIGH	3	$7.26	$8.17	+12.5%	+9.1
2025-09-15	RAPP	HIGH	3	$24.73	$29.60	+19.7%	+16.4
2025-09-15	TTGT	MEDIUM	2	$6.08	$5.20	-14.5%	-17.8
2025-09-04	SBH	HIGH	4	$14.13	$15.39	+8.9%	+3.3
2025-09-02	FTRE	MEDIUM	2	$10.01	$12.44	+24.3%	+17.7

August 2025

n = 15 · hit rate 60% · mean return +7.2% · mean alpha +0.4 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-08-27	ICUI	MEDIUM	2	$126.62	$151.90	+20.0%	+15.3
2025-08-19	PLMR	MEDIUM	2	$120.86	$128.48	+6.3%	+2.0
2025-08-15	KRO	HIGH	6	$5.84	$4.72	-19.2%	-23.9
2025-08-14	AAOI	HIGH	3	$21.01	$23.94	+13.9%	+7.7
2025-08-14	ESTA	MEDIUM	2	$37.95	$63.19	+66.5%	+60.3
2025-08-13	PRGS	HIGH	3	$46.11	$43.09	-6.5%	-12.8
2025-08-13	TROX	HIGH	6	$3.66	$3.45	-5.7%	-11.9

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-08-11	TNDM	HIGH	2	$10.11	$16.37	+61.9%	+54.5
2025-08-11	PTLO	HIGH	4	$7.85	$4.73	-39.8%	-47.2
2025-08-08	AUB	MEDIUM	2	$31.39	$31.70	+1.0%	-4.5
2025-08-06	BRBR	HIGH	3	$38.95	$30.33	-22.1%	-29.1
2025-08-06	TFX	HIGH	3	$111.09	$123.66	+11.3%	+4.3
2025-08-05	OI	HIGH	3	$13.00	$11.76	-9.5%	-18.7
2025-08-05	PRLB	MEDIUM	2	$44.20	$53.47	+21.0%	+11.8
2025-08-01	CNOB	MEDIUM	2	$21.77	$23.58	+8.3%	-1.3

July 2025

n = 2 · hit rate 50% · mean return +63.5% · mean alpha +55.8 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-07-31	TLRY	HIGH	3	$5.80	$13.60	+134.5%	+125.4
2025-07-16	HELE	HIGH	2	$22.39	$20.73	-7.4%	-13.8

June 2025

n = 10 · hit rate 90% · mean return +40.1% · mean alpha +31.1 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-06-26	PVH	MEDIUM	2	$64.98	$89.43	+37.6%	+29.3
2025-06-24	GIII	MEDIUM	2	$21.86	$26.42	+20.9%	+10.7
2025-06-23	SVRA	MEDIUM	2	$2.16	$3.49	+61.6%	+50.1
2025-06-18	NUVB	HIGH	2	$1.79	$3.27	+82.7%	+71.9
2025-06-10	CALY	HIGH	4	$7.70	$9.44	+22.6%	+14.7
2025-06-06	MATV	HIGH	5	$6.01	$11.82	+96.7%	+88.0
2025-06-05	MAGN	HIGH	7	$13.04	$11.61	-11.0%	-19.8
2025-06-05	IOVA	HIGH	3	$1.80	$2.21	+22.8%	+13.9
2025-06-04	STGW	HIGH	4	$4.27	$5.59	+30.9%	+23.1
2025-06-04	OFIX	MEDIUM	2	$10.90	$14.89	+36.6%	+28.9

May 2025

n = 15 · hit rate 73% · mean return +24.0% · mean alpha +14.4 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-05-30	INNV	MEDIUM	2	$4.09	$3.83	-6.4%	-16.8

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-05-23	VFC	MEDIUM	2	$11.73	$12.82	+9.3%	-0.8
2025-05-23	PHAT	MEDIUM	2	$4.02	$10.95	+172.4%	+162.3
2025-05-19	SNDX	HIGH	3	$9.30	$15.87	+70.7%	+62.2
2025-05-19	NPKI	MEDIUM	2	$8.01	$9.92	+23.9%	+15.4
2025-05-19	VYX	MEDIUM	2	$10.58	$12.85	+21.5%	+13.0
2025-05-15	RPAY	HIGH	2	$3.84	$5.70	+48.4%	+38.9
2025-05-14	BNL	MEDIUM	2	$14.77	$15.87	+7.5%	-2.2
2025-05-14	ERII	MEDIUM	2	$12.31	$14.30	+16.2%	+6.5
2025-05-14	AESI	MEDIUM	2	$12.72	$11.58	-9.0%	-18.7
2025-05-13	THRM	HIGH	3	$28.32	$32.69	+15.4%	+6.8
2025-05-13	PRG	MEDIUM	2	$28.28	$31.80	+12.4%	+3.8
2025-05-12	COLD	HIGH	2	$16.63	$13.56	-18.5%	-27.9
2025-05-12	NEOG	HIGH	8	$6.50	$5.11	-21.4%	-30.8
2025-05-05	PDM	MEDIUM	2	$6.49	$7.62	+17.4%	+5.1

April 2025

n = 6 · hit rate 83% · mean return +14.1% · mean alpha -3.7 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-04-25	CFFN	HIGH	3	$5.37	$5.97	+11.2%	-4.4
2025-04-17	CVGW	HIGH	3	$25.54	$25.75	+0.8%	-18.1
2025-04-15	HUMA	HIGH	5	$1.54	$2.47	+60.4%	+43.8
2025-04-14	REPX	MEDIUM	2	$22.60	$26.08	+15.4%	-0.8
2025-04-14	RCKT	MEDIUM	2	$5.91	$3.05	-48.4%	-64.6
2025-04-07	TITN	MEDIUM	2	$14.59	$21.22	+45.4%	+22.0

March 2025

n = 23 · hit rate 65% · mean return +12.0% · mean alpha +5.4 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-03-31	AAP	HIGH	4	$38.15	$45.59	+19.5%	+8.7
2025-03-21	WBTN	MEDIUM	2	$8.73	$8.36	-4.2%	-9.9
2025-03-19	CRGY†	MEDIUM	2	$11.16	$9.24	-17.2%	-22.9
2025-03-18	NVRI	HIGH	2	$6.56	$8.51	+29.7%	+22.0
2025-03-17	FNKO	MEDIUM	2	$7.19	$5.07	-29.5%	-36.1

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-03-17	DK	HIGH	4	$15.35	$21.58	+40.6%	+34.0
2025-03-17	MAC	MEDIUM	2	$16.47	$15.63	-5.1%	-11.7
2025-03-17	HRTG	MEDIUM	2	$12.79	$23.41	+83.0%	+76.5
2025-03-13	NFE	HIGH	2	$8.51	$3.20	-62.4%	-71.8
2025-03-13	NE	HIGH	3	$21.10	$27.92	+32.3%	+22.9
2025-03-11	MFIC	MEDIUM	2	$11.00	$11.59	+5.4%	-2.8
2025-03-11	STRA	HIGH	4	$77.69	$83.61	+7.6%	-0.6
2025-03-11	AMRC	HIGH	4	$10.32	$15.97	+54.8%	+46.5
2025-03-11	IE	HIGH	2	$5.71	$8.27	+44.8%	+36.6
2025-03-11	UCTT	HIGH	3	$22.98	$21.34	-7.1%	-15.3
2025-03-10	ACA	MEDIUM	2	$78.38	$89.80	+14.6%	+7.3
2025-03-10	MITK	HIGH	4	$8.83	$10.34	+17.1%	+9.8
2025-03-10	EVH	HIGH	4	$9.10	$8.48	-6.8%	-14.1
2025-03-06	ECPG	HIGH	3	$35.84	$38.29	+6.8%	+2.5
2025-03-06	JELD	MEDIUM	2	$5.92	$3.83	-35.3%	-39.7
2025-03-05	TALK†	HIGH	3	$2.76	$3.41	+23.6%	+21.0
2025-03-05	FTDR	MEDIUM	2	$42.05	$56.84	+35.2%	+32.6
2025-03-03	TBLA	MEDIUM	2	$2.81	$3.64	+29.5%	+27.7

February 2025

n = 6 · hit rate 33% · mean return +3.9% · mean alpha +5.1 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2025-02-28	MBC	MEDIUM	2	$13.99	$10.27	-26.6%	-26.2
2025-02-27	RCUS	MEDIUM	2	$9.88	$9.13	-7.6%	-8.3
2025-02-27	PPTA	HIGH	5	$8.26	$14.14	+71.2%	+70.4
2025-02-25	ECG	HIGH	3	$42.30	$59.45	+40.5%	+40.8
2025-02-19	HP	HIGH	3	$24.61	$15.63	-36.5%	-33.5
2025-02-03	ASH	HIGH	2	$58.70	$48.22	-17.9%	-12.4

December 2024

n = 4 · hit rate 50% · mean return +10.6% · mean alpha +17.0 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2024-12-26	PGNY	MEDIUM	2	$14.71	$21.84	+48.5%	+53.6

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2024-12-17	XPRO	MEDIUM	2	$11.09	$10.59	-4.5%	+1.3
2024-12-16	HRZN	MEDIUM	2	$6.85	$7.66	+11.8%	+18.0
2024-12-11	TCNNF	HIGH	10	$4.83	$4.19	-13.2%	-5.1

November 2024

n = 1 · hit rate 100% · mean return +6.8% · mean alpha +7.3 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2024-11-27	TCPC	MEDIUM	2	$7.16	$7.65	+6.8%	+7.3

September 2024

n = 1 · hit rate 100% · mean return +101.2% · mean alpha +88.6 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2024-09-06	UEC	MEDIUM	2	$4.26	$8.57	+101.2%	+88.6

August 2024

n = 7 · hit rate 100% · mean return +27.6% · mean alpha +18.2 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2024-08-23	ATEC	HIGH	4	$5.74	$10.21	+77.9%	+71.9
2024-08-22	CSV	MEDIUM	2	$30.84	$37.75	+22.4%	+15.9
2024-08-16	PCRX	HIGH	5	$12.94	$16.37	+26.5%	+19.1
2024-08-13	NVST	HIGH	2	$16.99	$20.75	+22.1%	+11.3
2024-08-13	EYE	MEDIUM	2	$10.61	$11.82	+11.4%	+0.6
2024-08-12	GT	HIGH	4	$7.78	$9.77	+25.6%	+12.9
2024-08-07	CTOS	HIGH	2	$4.05	$4.35	+7.4%	-4.1

June 2024

n = 3 · hit rate 67% · mean return +1.9% · mean alpha -0.9 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2024-06-17	BOC	HIGH	7	$13.90	$14.74	+6.0%	+2.8
2024-06-11	ONDS	MEDIUM	2	$0.86	$0.89	+3.5%	+1.4
2024-06-07	LEG	HIGH	6	$11.77	$11.32	-3.8%	-7.1

May 2024

n = 5 · hit rate 40% · mean return -4.6% · mean alpha -9.2 pp

Filing Date	Ticker	Conv	#I	Entry	Exit	90d Return	Alpha (pp)
2024-05-30	BCRX	HIGH	9	$6.57	$8.55	+30.1%	+23.0
2024-05-29	PRVA	MEDIUM	2	$16.40	$20.76	+26.6%	+19.5
2024-05-16	ASTE	MEDIUM	2	$32.81	$31.42	-4.2%	-7.4
2024-05-15	DNA	MEDIUM	2	$33.56	$11.24	-66.5%	-69.2
2024-05-13	CNMD	HIGH	4	$71.42	$64.97	-9.0%	-11.7

† Signal with contemporaneous hostile Schedule 13D filing (3 of 141).

The same data is exported in the CSV and Parquet appendix for programmatic ingestion — see Section 7 for the file index.

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6. Academic Foundation

The insider open-market purchase signal is not novel. It rests on three decades of peer-reviewed research. LookInsight's contribution is not the thesis — the thesis has been published, tested, and replicated. LookInsight's contribution is the production-scale engineering that makes the thesis deliverable as a clean alpha signal for a systematic desk.

Key references in the foundation:

- **Lakonishok, J. & Lee, I. (2001)** — "*Are Insider Trades Informative?*" Review of Financial Studies, 14(1). Large-sample NYSE/AMEX/NASDAQ evidence of abnormal returns following insider purchases, stronger in smaller firms.
- **Jeng, L., Metrick, A. & Zeckhauser, R. (2003)** — "*Estimating the Returns to Insider Trading.*" Review of Financial Studies, 16(2). Documented that insider purchases earn approximately 11% abnormal annual returns, with strong asymmetry between purchases and sells.
- **Cohen, L., Malloy, C. & Pomorski, L. (2012)** — "*Decoding Inside Information.*" Journal of Finance, 67(3). Separated opportunistic from routine insider trading; opportunistic insider buys predict future returns far more strongly than routine trades — the academic anchor for filtering GENUINE P from scheduled P.
- **Brav, A., Jiang, W., Partnoy, F. & Thomas, R. (2008)** — "*Hedge Fund Activism, Corporate Governance, and Firm Performance.*" Journal of Finance, 63(4). Return dynamics around Schedule 13D filings — the foundation for the activist overlay exposed via `hostile_flag`.

What the academic literature leaves to an implementer:

- **Structured classification of Form 4 transactions.** Not every `P` code is a genuine conviction purchase. Removing structured placements, employee stock-purchase programs, option-related P codes, and multi-entity duplication is a prerequisite that the literature assumes, not performs.

- **Temporal clustering across multiple insiders.** Canonical insider-alpha studies are single-insider by default. Extending to multi-insider clusters with rolling windows, shared-name aggregation, and cluster-value aggregation is an engineering construction that does not exist off the shelf.
- **Historical market cap estimation and size-filtered cohorts.** Required to reproduce the midcap cut that our own data confirms as the strongest band.
- **Filing-date anchoring.** The literature often measures returns from transaction date. Filing-date anchoring eliminates look-ahead bias and matches what a third party could actually trade — a small but load-bearing distinction.
- **Production-scale maturity bookkeeping and survivorship-aware return attribution.**

Each of those steps is months of data engineering per team. LookInsight has productionized them, exposed the output in the 33-column data appendix, and documented the methodology in this brief. A quantitative desk evaluating LookInsight is not being asked to validate the underlying thesis — the thesis is literature. The desk is being asked to verify that the engineering faithfully translates that literature into actionable signals — and the data appendix (every row tied to its SEC EDGAR filing URL) makes that verification a one-click exercise.

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7. Appendix

File index

File	Purpose
`signals_v1_1_2026-04-19.csv`	141-row signal export with 33-column schema (every row carries its SEC EDGAR filing URL)
`signals_v1_1_2026-04-19.parquet`	Same data, typed Parquet for pandas/polars/Spark
`DATA_DICTIONARY.md`	Authoritative column specification and derivation formulas
`brief_v1_1.md`	This document
`stats.json`	Machine-readable version of every number cited in this brief
`charts/funnel.png`	Filter funnel visualization
`charts/return_distribution.png`	90-day return histogram
`charts/alpha_waterfall.png`	Signal alpha distribution

Glossary

- **Strong_buy:** the cohort definition — a cluster satisfying all of: direction = buy, ≥2 insiders, ≥$100K total value, midcap market cap, earnings proximity ≤ 60 days.
- **Conviction tier:** classification of a cluster's strength (strong_buy, buy, watch). This brief covers only strong_buy.

- **Wilson CI:** a confidence interval for a binomial proportion that has better small-sample coverage than the normal-approximation interval.
- **Alpha (90d):** a signal's 90-day return minus the SPY 90-day return over the same calendar window, expressed in percentage points.
- **Hostile flag:** true if the issuer had a contemporaneous Schedule 13D filing containing hostile-activist language. Informational only — not a cohort filter.
- **Filing-date return:** return measured starting from the close on the SEC filing date, not the underlying transaction date. Avoids look-ahead bias.

LookInsight research brief. Cohort: 141 mature strong_buy signals. Snapshot: April 19, 2026. *Manohar · manohar@lookinsight.ai · ci.lookinsight.ai* *Prepared for institutional review.*